

ETF Momentum Research

Question: does volatility scaling and an SPY 200-day moving-average filter improve long-only 12-1 ETF momentum after realistic transaction costs?

Research design

The study uses 24 liquid US-listed ETFs across equities, rates, credit, commodities, FX, and sectors. Signals use returns from 252 to 21 trading days before month-end. Portfolio weights become active on the following trading day. The research period is 2011-2018; the out-of-sample period is 2019 through 31 August 2026. Each rebalance applies 10 bps times one-way turnover.

Strategies

The benchmark holds all available ETFs equally. The three active variants hold the six strongest 12-1 signals with equal weights, inverse-volatility weights, or inverse-volatility weights only when SPY is above its 200-day moving average. The filter otherwise holds cash.

Research-period results

Strategy	CAGR	Vol	Sharpe	Max DD	Annual turnover
Equal weight	5.4%	9.9%	0.58	-14.8%	0.1x
12-1 momentum	6.2%	11.5%	0.58	-23.8%	4.6x
Momentum inverse vol	6.6%	10.9%	0.64	-19.5%	5.7x
Momentum regime filter	7.0%	10.2%	0.71	-18.2%	5.5x

Interpretation

During the research period, the regime-filtered variant produced the highest reported Sharpe ratio in this fixed design. That pattern is a hypothesis, not a trading conclusion, because the design still reflects a chosen universe and model parameters.

Out-of-sample results

Strategy	CAGR	Vol	Sharpe	Max DD	Annual turnover
Equal weight	12.9%	12.8%	1.01	-26.9%	0.0x
12-1 momentum	17.6%	17.2%	1.03	-24.5%	5.1x
Momentum inverse vol	15.5%	15.0%	1.04	-23.2%	7.2x
Momentum regime filter	8.3%	12.2%	0.71	-17.9%	6.9x

Key observation

The out-of-sample results do not support a simple claim that the regime filter improves the strategy. In this run, inverse-volatility momentum delivered the highest out-of-sample Sharpe ratio, while the filter lowered volatility and drawdown but also lowered CAGR. This is the kind of result that should change the research narrative rather than be omitted.

What the project does well

The implementation separates research and out-of-sample periods, shifts weights by one trading day, records turnover, applies explicit costs, and writes all weights and daily returns to CSV. Unit tests check signal timing, weight constraints, cost application, and drawdown metrics.

Limitations

The ETF universe is selected with hindsight, adjusted-close data can differ across providers, monthly close execution is stylised, and the strategy has no borrow, tax, market-impact, capacity, or point-in-time constituent model. Parameter sensitivity and alternative universes are necessary before treating the result as robust.

Reproducibility

Run ``python3 src/run_research.py --download --end 2026-08-31`` in ``momentum_research/``. The downloader uses the public Yahoo Finance chart endpoint and caches raw CSVs locally. This report is an educational research artifact, not investment advice.